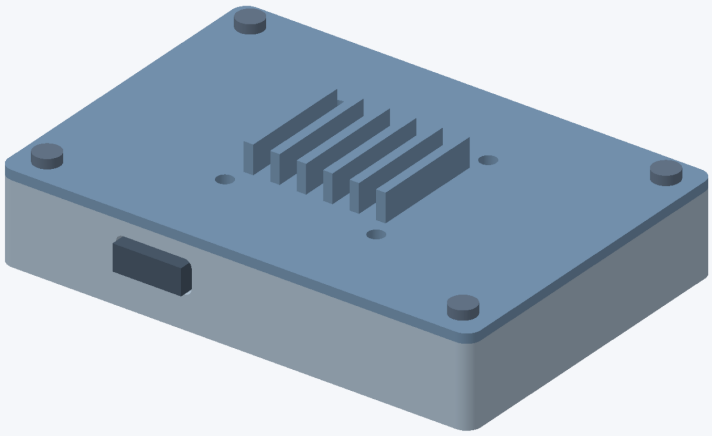
DIGITAL MECHANICAL DESIGN

A compact enclosure with a thermal lid.

Compact electronics enclosure



Digital design study | Rev C | CAD geometry

100 × 70 × 25 mm	Two custom enclosure parts	8 W thermal study
Parametric passive housing	Tray and removable lid	Passive limit exposed

What I designed. An independent generic compute enclosure with removable finned aluminum lid, integral thermal pedestal, CNC/print plans, dimensioned interfaces, an inspection fixture and rerunnable validation.

Engineering judgment. The assumed passive case reaches 93.4°C at 35°C ambient, missing an 80°C target. A fan hypothesis reaches 61.4°C but needs testing. The 0.31–0.69 mm pad-gap stack makes supplier compression data and real board measurements the next design inputs.

Evidence. 31 component export pairs checked; 38 tests executed, 0 failures; 361 assembly pair checks.

Digital comparison. Three fin alternatives were rebuilt. Taller fins add area but exceed the height target; eight fins violate the selected cutter-width constraint. Cooling remains conditional on unverified heat-transfer assumptions.

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Digital design and manufacturing planning. No completed fabrication or physical test is claimed.

github.com/gokulg846/compact-edge-enclosure